1. Write a program to perform Linear Search.

```
#include <stdio.h>
int main() {
    int n, i, key;
    scanf("%d", &n);
    int arr[n];
    for(i=0;i<n;i++) scanf("%d",&arr[i]);
    scanf("%d",&key);
    for(i=0;i<n;i++){
        if(arr[i]==key){
            printf("Found at %d", i+1);
            return 0;
        }
    }
    printf("Not Found");
}
```

2. Write a program to perform Binary Search.

```
#include <stdio.h>
int main() {
    int n, i, key, low=0, high, mid;
    scanf("%d",&n);
    int arr[n];
    for(i=0;i<n;i++) scanf("%d",&arr[i]);
    scanf("%d",&key);
    high=n-1;
    while(low<=high){
        mid=(low+high)/2;
        if(arr[mid]==key){
            printf("Found at %d",mid+1);
            return 0;
        }
        else if(arr[mid]<key) low=mid+1;
        else high=mid-1;
    }
    printf("Not Found");
}
```

3. Write a program to implement Stack using array.

```
#include <stdio.h>
#define MAX 5
```

```
int stack[MAX], top=-1;
void push(int x){
    if(top==MAX-1) printf("Overflow");
    else stack[++top]=x;
}
void pop(){
    if(top==-1) printf("Underflow");
    else printf("%d", stack[top--]);
}
int main(){
    push(10);

    push(20);

    pop();
}
```

4. Write a program to implement Queue using array.

```
#include <stdio.h>
#define MAX 5
int q[MAX], front=-1, rear=-1;
void enqueue(int x){
    if(rear==MAX-1) printf("Overflow");
    else{
        if(front==-1) front=0;
        q[++rear]=x;
    }
}
void dequeue(){
    if(front==-1 || front>rear) printf("Underflow");
    else printf("%d", q[front++]);
}
int main(){
    enqueue(10); enqueue(20); dequeue();
}
```

5. Write a program to create and display a linked list.

```
#include <stdio.h>
#include <stdlib.h>
struct node{
    int data;
    struct node *next;
};
```

```
struct node *head=NULL;
void create(int n){
    struct node *temp,*newnode;
    for(int i=0;i<n;i++){
        newnode=(struct node*)malloc(sizeof(struct node));
        scanf("%d",&newnode->data);
        newnode->next=NULL;
        if(head==NULL) head=temp=newnode;
        else{
            temp->next=newnode;
            temp=newnode;
        }
    }
}
void display(){
    struct node *temp=head;
    while(temp!=NULL){
        printf("%d->",temp->data);
        temp=temp->next;
    }
}
int main(){
    int n; scanf("%d",&n);
    create(n);
    display();
}
```

6. Write a program to reverse an array.

```
#include <stdio.h>
int main() {
    int n, i;
    scanf("%d", &n);
    int arr[n];
    for(i=0;i<n;i++) scanf("%d",&arr[i]);
    for(i=n-1;i>=0;i--) printf("%d ", arr[i]);
}
```

7. Write a program to find maximum and minimum in an array.

```
#include <stdio.h>
int main() {
    int n, i, max, min;
    scanf("%d",&n);
    int arr[n];
```

```
    for(i=0;i<n;i++) scanf("%d",&arr[i]);
    max = min = arr[0];
    for(i=1;i<n;i++){
        if(arr[i]>max) max=arr[i];
        if(arr[i]<min) min=arr[i];
    }
    printf("Max=%d Min=%d", max, min);
}
```

8. Write a recursive program to find factorial of a number.

```
#include <stdio.h>
int fact(int n){
    if(n==0 || n==1)
        return 1;
    else
        return n * fact(n-1);
}
int main(){
    int n;
    scanf("%d",&n);
    printf("%d", fact(n));
}
```

9. Write a recursive program to generate Fibonacci series.

```
#include <stdio.h>
int fib(int n){
    if(n==0) return 0;
    if(n==1) return 1;
    return fib(n-1) + fib(n-2);
}
int main(){
    int n,i;
    scanf("%d",&n);
    for(i=0;i<n;i++)
        printf("%d ", fib(i));
}
```

10. Write a recursive program to find sum of digits of a number.

```
#include <stdio.h>
int sum(int n){
    if(n==0)
        return 0;
```

```
    else
        return (n%10) + sum(n/10);
}
int main(){
    int n;
    scanf("%d",&n);
    printf("%d", sum(n));
}
```

11. Write a recursive program to reverse a number.

```
#include <stdio.h>
int rev(int n, int r){
    if(n==0)
        return r;
    else
        return rev(n/10, r*10 + n%10);
}
int main(){
    int n;
    scanf("%d",&n);
    printf("%d", rev(n,0));
}
```

12. Write a recursive program to find power of a number.

```
#include <stdio.h>
int power(int a, int b){
    if(b==0)
        return 1;
    else
        return a * power(a, b-1);
}
int main(){
    int a,b;
    scanf("%d %d",&a,&b);
    printf("%d", power(a,b));
}
```